

Homework 11

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Problem 2.1. Prove the Brouwer fixed point theorem for maps $f : D^n \rightarrow D^n$ by applying degree theory to the map $S^n \rightarrow S^n$ that sends both the northern and southern hemispheres of S^n to the southern hemisphere via f . [This was Brouwer's original proof.]

Proof. $\forall f : D^n \rightarrow D^n$, let $f' : S^n \rightarrow S^n$ where $(\vec{x}, y) \rightarrow (f(\vec{x}), -|y|)$. $\deg f' = 0$ since f' is not onto. Thus $\exists a = (\vec{x}, y) \in S^n$ where $\vec{x} \in D^n$ and $y \in \mathbb{R}$ such that $|\vec{x}|^2 + |y|^2 = 1$ such that $f'(a) = a$. Thus $(\vec{x}, y) = a = f'(a) = (f(\vec{x}), -|y|)$. Thus $y = -|y| \implies y \leq 0$ and $\vec{x} = f(\vec{x})$. Thus f has a fixed point $\vec{x}$. $\square$

Problem 2.2. Given a map $f : S^{2n} \rightarrow S^{2n}$, show that there is some point $x \in S^{2n}$ with either $f(x) = x$ or $f(x) = -x$. Deduce that every map $\mathbb{R}P^{2n} \rightarrow \mathbb{R}P^{2n}$ has a fixed point. Construct maps $\mathbb{R}P^{2n-1} \rightarrow \mathbb{R}P^{2n-1}$ without fixed points from linear transformations $\mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ without eigenvectors.

Proof. It suffices to assume that f doesn't have a fixed point. Then $\deg(f) = (-1)^{2n+1} = -1$. Then $\deg(-f) = \deg(\text{antipodal}) \deg(f) = (-1)^{2n+1} * (-1) = 1 \implies -f$ has a fixed point. Thus $\exists x \in S^{2n} \ni x = (-f)(x) = -f(x) \implies f(x) = -x$.

Let $q : S^{2n} \rightarrow \mathbb{R}P^{2n}$ be standard quotient map. $p : S^{2n} \rightarrow \mathbb{R}P^{2n}$ is the universal cover of $\mathbb{R}P^{2n}$. $\forall f : \mathbb{R}P^{2n} \rightarrow \mathbb{R}P^{2n}$. By the Lifting Criterion, $f \circ q$ lifts to $g : S^{2n} \rightarrow S^{2n} \ni p \circ g = f \circ q$.

$$\begin{array}{ccccc} & & & & S^{2n} \\ & & & \nearrow g & \\ S^{2n} & \xrightarrow{q} & \mathbb{R}P^{2n} & \xrightarrow{f} & \mathbb{R}P^{2n} \\ & & & \searrow p & \end{array}$$

By the previous argument, $\exists x \in S^{2n} \ni g(x) = \pm x$. Thus $[x] = (p \circ g)(x) = (f \circ q)(x) = f([x])$. Thus $[x]$ is a fixed point of f .

Let $A = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$ where I_n is the $n \times n$ identity matrix. The eigenvalues of A are $\pm i$ and thus A has no real eigenvectors. So the the 1-dimensional subspaces of $\mathbb{R}^{2n}$ are never fixed by A . $\mathbb{R}P^{2n-1} = \mathbb{R}^{2n}/(x \sim \lambda x, \forall \lambda \in \mathbb{R}/\{0\})$ or the set of all one dimensional subspaces of $\mathbb{R}^{2n}$. Thus no point of $\mathbb{R}P^{2n-1}$ is fixed by A . $\square$

Problem 2.4. Construct a surjective map of $S^n \rightarrow S^n$ with degree 0 for $n \geq 1$

Proof. Let $\omega : S^1 \rightarrow S^1$ be the identity loop, $x \rightarrow x$. Let $f_1 = \omega \cdot \bar{\omega}$, ie go completely counterclockwise through S^1 , then go completely clockwise through S^1 . f_1 is surjective. If you think about $S^1 \subseteq \mathbb{C}$, $f_1^{-1}(i) = \{e^{i\pi/4}, e^{7i\pi/4}\}$, $\deg(f_1)|_{e^{i\pi/4}} = 1$ and $\deg(f_1)|_{e^{7i\pi/4}} = -1$. By Prop 2.30, $\deg(f_1) = 0$.

Let $f_n := S^{n-1}f_1$ where S^n is the n th suspension of f_1 . Since $\deg Sg = \deg g, \forall g : S^n \rightarrow S^n \implies \deg S^n g = \deg g$. Thus $\deg f_n = \deg f = 0$. $\square$

Claim 2.7A. Let A be a matrix of the following form:

$$A_{x,y} = \begin{cases} 1 & \text{if } x = y \neq i \\ \alpha & \text{if } x = y = i \\ \beta & \text{if } x = i \text{ and } y = j \\ 0 & \text{otherwise} \end{cases}$$

for some $i, j \in [0, \dots, n] \ni i \neq j$ for $\alpha > 0$ and for $\beta \in \mathbb{R}$, $\det(A) = \alpha$

Proof. If $\beta = 0$, A is a diagonal matrix with entries equal to 1 on the diagonal, except for $A_{i,i} = \alpha$. In this case, the determinant is simply the product of the diagonal entries, so

$$\det(A) = \alpha \cdot 1 \cdot \dots \cdot 1 = \alpha.$$

Now, consider the case when $\beta \neq 0$. Let $A_{(x,y)}$ be the submatrix formed from A by removing the x th row and y th column. To simplify, we perform a cofactor expansion along row i , which contains both $A_{i,i} = \alpha$ and $A_{i,j} = \beta$:

$$\det(A) = \alpha \det(A_{(i,i)}) \pm \beta \det(A_{(i,j)})$$

Note $A_{(i,i)} = I$. Thus $\det(A_{(i,i)}) = 1$. Furthermore, when you remove the j th column, then you remove $a_{j,j}$ and thus the row corresponding to the j -th row in $A_{i,j}$ has no nonzero elements as the only nonzero element in the row is $a_{j,j}$. Thus $\det(A_{(i,j)}) = 0$. Thus $\det(A) = \alpha$. $\square$

Claim 2.7B. If R is the invertible matrix corresponding to the row operations $ar_i + br_j \rightarrow r_i$, for $a > 0, b \in \mathbb{R}$, then $\exists$ a path $S : [0, 1] \rightarrow GL_n(\mathbb{R})$ from R to I in $GL_n(\mathbb{R})$.

Proof. Suppose $R \in GL_n(\mathbb{R})$ is the invertible matrix corresponding to the row operation $ar_i + br_j \rightarrow r_i$ for $a > 0$ and $b \in \mathbb{R}$. Thus

$$R_{x,y} = \begin{cases} 1 & \text{if } x = y \neq i \\ a & \text{if } x = y = i \\ b & \text{if } x = i \text{ and } y = j \\ 0 & \text{otherwise} \end{cases}$$

By Claim 2.7A, $\det(R) = a > 0$. Define,

$$(S_t)_{x,y} = \begin{cases} 1 & \text{if } x = y \neq i \\ t(a-1) + 1 & \text{if } x = y = i \\ tb & \text{if } x = i \text{ and } y = j \\ 0 & \text{otherwise} \end{cases}$$

For all $t \in [0, 1]$, by Claim 2.7A $\det(S_t) = t(a-1) + 1 > 0$. $S_0 = I$ and $S_1 = R$, thus $\exists$ a path from R to I in $GL_n(\mathbb{R})$. $\square$

Claim 2.7C. $\forall A \in GL_n(\mathbb{R}), \exists D = \text{diag}(\pm 1, \dots, \pm 1) \ni A \simeq D$ and $\det(D) = \frac{\det(A)}{|\det(A)|}$

Proof. $\forall A \in GL_n(\mathbb{R})$, you basically want to use Gaussian Elimination to come up with a series of Row operation matrices like the ones above.

Let $i = 0$. Let $m = 0$;

1. While $i \leq n - 1$:

(a) If $a_{i,i} = 0, \exists y \in [0, \dots, n - 1] \ni a_{r,i} \neq 0$. Let R_m be the matrix which does $r_i + r_y \rightarrow r_i$. Then $m := m + 1$.

(b) Let R_m be the row operation which does $\frac{1}{|a_{i,i}|} r_i \rightarrow r_i$. $m := m + 1$. Thus $a_{i,i} = \pm 1$.

(c) Let $j = 0$. While $j \leq n - 1$:

i. If $i \neq j$ and $a_{j,i} \neq 0$: Let $x = a_{j,i}$. If $a_{i,i} > 0, x = -a_{j,i}$. Let R_m be the row operation which does $xr_i + yr_j \rightarrow r_j$. Thus, $a_{j,i} = 0$. $m := m + 1$.

(d) $i := i + 1$.

2. Now $A \prod_{i=0}^{m-1} R_i$ is a diagonal matrix with ± 1 on the diagonal. Let $i = n - 1$.

By Claim 2.7B, $\forall i \in [0, \dots, m - 1] \exists S_i : [0, 1] \rightarrow GL_n(\mathbb{R})$ a path from I to R_i where $\forall t \in [0, 1], S_i(t) > 0$. Thus $H_t : \mathbb{R}^n \rightarrow \mathbb{R}^n$ where $\vec{x} \rightarrow A \prod_{i=1}^{m-1} S_i(t) \vec{x}$ where $H_0 = A \prod_{i=1}^{m-1} S_i(0) = A$ since $S_i(0) = I$ and $H_1 = A \prod_{i=1}^{m-1} S_i(1) = A \prod_{i=1}^{m-1} R_i = D$ where D is a diagonal matrix with ± 1 on diagonal. Thus $A \simeq D$.

Furthermore since $\det(R_i) > 0$, $\det(D)$ is positive if and only if $\det(A)$ is positive. Furthermore since D is a diagonal matrix with plus and minuses, $\det(D) = \frac{\det(A)}{|\det(A)|}$. $\square$

Problem 2.7. For an invertible linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ show that the induced map on $H_n(\mathbb{R}^n, \mathbb{R}^n - 0) \simeq \tilde{H}_{n-1}(\mathbb{R}^n - 0) \cong \mathbb{Z}$ is 1 or -1 according to whether the determinant of f is positive or negative.

Proof. $\forall A \in GL_n(\mathbb{R})$,

by Claim 2.7C $\exists (a_0, \dots, a_{n-1}) \in \{0, 1\}^n$ such that $A \simeq D = \text{diag}((-1)^{a_0}, \dots, (-1)^{a_n})$.

Let $m = |(a_0, \dots, a_{n-1})|^2$, D does m reflections to $\mathbb{R}^n - \{0\} \simeq S^{n-1}$. Furthermore, $\det D = \prod_{i=1}^{n-1} (-1)^{a_i} = (-1)^m = \frac{\det(A)}{|\det(A)|}$. $\deg(A) = \deg(D) = (-1)^m = \det D = \frac{\det(A)}{|\det(A)|}$. So if $\det(A) > 0 \implies \deg(A) > 0 \implies \deg(A) = 1$ and similarly if $\det(A) < 0 \implies \deg(A) < 0 \implies \deg(A) = -1$. So if the determinant of A is positive the induced map is 1 otherwise its -1. $\square$

Problem 2.8. $\forall f \in \mathbb{C}[x]$, f can be extended to a map of one point compactifications $\hat{f} : S^2 \rightarrow S^2$. Show $\deg \hat{f} = \deg f$. Show $\forall x \in \{\text{roots of } f\}, \deg \hat{f}|_x = \text{mult}(x)$

Proof. $\forall f \in \mathbb{C}[x]$, $f : \mathbb{C} \rightarrow \mathbb{C}$ where $x \rightarrow \sum_{i=0}^n a_i x^i$ and $a_n \neq 0$. Let $H : \mathbb{C} \times I \rightarrow \mathbb{C}$ where $x \rightarrow a_n x^n + t \left(\sum_{i=0}^{n-1} a_i x^i \right)$. Thus $f \simeq [x \rightarrow a_n x^n]$. We need to check that $\hat{H} : S^2 \times I \rightarrow S^2$ is continous. It is continuous on $\mathbb{C} \times I$ as it just extends H , so we just need to check that its continous on $\{\infty\} \times I$. $\forall t \in I, \hat{H}(\infty, t) = \infty$. $\forall A > 0, \exists B > 0$ and $\epsilon > 0$ such that

$$|z| > B, |t_0 - t| > \epsilon \implies |H(z, t)| > A$$

This is true since $a_n z^n$ dominates H . Thus $\hat{H}$ is continous. Thus $\hat{f} \simeq [x \rightarrow \hat{a}_n x^n]$.

Let $a_n = re^{i\theta}$ where $r \neq 0$. Take $J : \mathbb{C} \times I \rightarrow \mathbb{C}$ where $x \rightarrow r^t e^{it\theta} x^n$. Thus $[x \rightarrow a_n x^n] \simeq [x \rightarrow x^n]$. $\hat{J}$ is continuous for the same reason as above. Thus $\hat{f} \simeq [x \rightarrow a_n x^n] \simeq [x \rightarrow x^n]$ as functions of $S^2 \rightarrow S^2$.

Let $g_n := x \rightarrow x^n$. Look at $\hat{g}^{-1}(1) = \{e^{2\pi i p/n} : p \in [0, \dots, n-1]\}$. Note near $e^{2\pi i p/n}$, $\hat{g}$ is a local homeomorphism stretching the area by n , this stretching can be eliminated by a deformation of $\hat{g}$ near $e^{2\pi i p/n}$. So $\deg \hat{g}_n|_{e^{2\pi i p/n}} = 1$. Thus $\deg(f) = \deg \hat{g}_n = n$.

Let $f(x) = \prod_i (x - x_i)^{m_i}$ where $m_i \in \mathbb{N}$ and $\sum_i m_i = n$ and $i \neq j \implies x_i \neq x_j$. Near x_i , $f \simeq a(x - x_i)^{m_i} \simeq (x - x_i)^{m_i}$, thus for sufficiently small circles, f will wrap them around themselves m_i times. Thus $\deg f|_{x_i} = m_i$. $\square$

Problem 2.10a. Let X be the quotient space of S^2 with the identifications $x \sim -x$ for $x \in S^1$ equatorial

Proof. X as a CW complex is $\mathbb{R}P^1 = S^1$ and 2 2-cells corresponding to the northern hemisphere and southern hemisphere. $\varphi_N : \delta D^2 \rightarrow \mathbb{R}P^1 = S^1$ sends $x \rightarrow x^2$ is the attaching map for the northern hemisphere. $\varphi_S : \delta D^2 \rightarrow \mathbb{R}P^1 = S^1$ sends $x \rightarrow x^{-2}$. We have 1 0 cell, 1 1 cell, and 2 2 cells.

$$0 \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$$

$$d_1 = 0$$

$$d_2 = \begin{bmatrix} 2 & -2 \end{bmatrix}$$

$$\text{Thus } H_0(X) = \mathbb{Z}, H_1(X) = \frac{\mathbb{Z}}{2\mathbb{Z}} = \mathbb{Z}_2, \text{ and } H_2(X) = \mathbb{Z}$$

$\square$

Problem 2.10b. Let X be the quotient space of S^3 with the identifications $x \sim -x$ for $x \in S^2$ equatorial

Proof. X as a CW complex is $\mathbb{R}P^2$ and 2 3-cells corresponding to the northern hemisphere and southern hemisphere. $\varphi_N : \delta D^3 \rightarrow \mathbb{R}P^2$ is the 2-covering. $\varphi_S = \varphi_N \circ (-1)$. We have 1 0 cell, 1 1 cell, 1 2-cells, and 2 3-cells.

$$0 \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$$

$$d_1 = 0$$

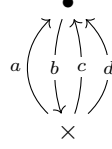
$$d_2 = 2$$

Note $d_3(\mathbb{Z}^2) \subseteq \ker d_2 = 0$ since $[x \rightarrow 2x]$ has no kernel, thus $\ker d_3 = \mathbb{Z}^2$. Thus $H_0(X) = \mathbb{Z}$, $H_1(X) = \frac{\mathbb{Z}}{2\mathbb{Z}} = \mathbb{Z}_2$, $H_2(X) = 0$ (since kernel of $[x \rightarrow 2x]$ is 0), $H_3(X) = \mathbb{Z}^2 / \{0\} = \mathbb{Z}^2$. $\square$

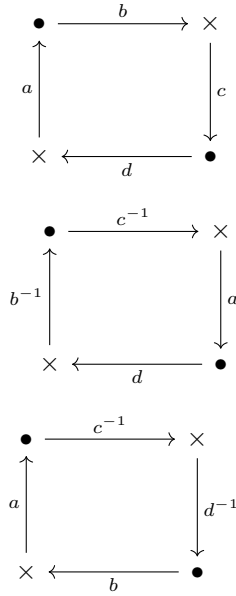
Problem 2.11. Let X be the space in 1.2.14. Compute $H_n(X)$.

Proof. X has two 0 cells, four 1 cells, three 2 cells, and one 3 cells.

The 1 skeleton looks like:



Here are the attaching maps for the 2-cells.



Thus the chain complex is:

$$0 \xrightarrow{0} \mathbb{Z} \xrightarrow{d_3} \mathbb{Z}^3 \xrightarrow{d_2} \mathbb{Z}^4 \xrightarrow{d_1} \mathbb{Z}^2 \xrightarrow{0} 0$$

$$d_1 = \begin{bmatrix} -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$d_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \\ 1 & 1 & -1 \end{bmatrix}$$

For d_3 we need to compute the degree's of the maps of δS^3 to each of the 2 cells. Let f be a two cell. $\varphi : \delta S^3 \rightarrow f$ is the attaching map. Take $\varphi^{-1}(x) = \{x_1, x_2\}$, where x_1 is on the front face, and x_2 where x is the center of f , φ is a local homeomorphism at x so $\deg \varphi|_{x_i} = \pm 1$. However x_1 and x_2 differ by a reflection and a rotation. Thus $\deg \varphi = 0$ onto each face. Thus $d_3 = 0$.

Doing row operations we see, $d_1 \rightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and $d_2 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

Thus

$$\begin{aligned}\ker d_1 &= \text{span}(e_1, e_2, e_3) \\ d_1(\mathbb{Z}^4) &= \text{span}(e_1) \\ \ker d_2 &= 0 \\ d_2(\mathbb{Z}^3) &= \text{span}(e_1, 2e_2, 2e_3)\end{aligned}$$

Thus

$$\begin{aligned}H_0(X) &= \frac{\mathbb{Z}^2}{\mathbb{Z}} = \mathbb{Z} \\ H_1(X) &= \frac{\text{span}(e_1, e_2, e_3)}{\text{span}(e_1, 2e_2, 2e_3)} = \mathbb{Z}_2 \oplus \mathbb{Z}_2 \\ H_2(X) &= 0 \\ H_3(X) &= \mathbb{Z}\end{aligned}$$

□

Problem 2.2.12a. Show that $q : S^1 \times S^1 \rightarrow S^2$, the quotient map, collapsing the $S^1 \vee S^1$ to a point, is not homotopic to a point by showing it induces an isomorphism on H_2 .

Proof. Look at the long exact sequence of $(S^1 \times S^1, S^1 \vee S^1)$.

□

Problem 2.2.14a. A map $f : S^n \rightarrow S^n$ is an even map if $f(x) = f(-x)$. Show that an even map must have even degree, and that degree must be 0 when n is even.

Proof. Let f be an even map. f factors through $\mathbb{R}P^n$, i.e. $\exists g : \mathbb{R}P^n \rightarrow S^n$ such that $f = gq$ where q is the standard quotient from $S^n \rightarrow \mathbb{R}P^n$. This induces maps of homology so we have:

$$\begin{array}{ccc} & & S^n \\ & \nearrow f & \uparrow g \\ S^n & \xrightarrow{q} & \mathbb{R}P^n \\ \\ & & H_n(S^n) \\ & \nearrow f_* & \uparrow g_* \\ H_n(S^n) & \xrightarrow{q_*} & H_n(\mathbb{R}P^n)\end{array}$$

1. If n is even: $H_n(\mathbb{R}P^n) = 0$, thus $q_0 = 0 \implies f_* = 0$ thus $\deg f_* = 0$.
2. If n is odd: We have the following long exact sequence from $(\mathbb{R}P^n, \mathbb{R}P^{n-1})$ and since $\mathbb{R}P^{n-1}$ is a skeleton of $\mathbb{R}P^n$ the pair is good. Moreover $\mathbb{R}P^n/\mathbb{R}P^{n-1} \cong S^n$. So there is an isomorphism between $H_n(\mathbb{R}P^n, \mathbb{R}P^{n-1}) \cong H_n(S^n)$ so we have,

$$0 = H_n(\mathbb{R}P^{n-1}) \longrightarrow H_n(\mathbb{R}P^n) \longrightarrow H_n(\mathbb{R}P^n, \mathbb{R}P^{n-1}) \cong H_n(S^n) \longrightarrow H_{n-1}(\mathbb{R}P^{n-1}) = 0$$

Since n is odd, $n-1$ is even thus $H_{n-1}(\mathbb{R}P^{n-1}) = 0$. Thus $H_n(\mathbb{R}P^n) \cong H_n(S^n) \cong \mathbb{Z}$, thus 1 maps to 1. Let $\varphi : \mathbb{R}P^n \rightarrow \mathbb{R}P^n/\mathbb{R}P^{n-1} = S^n$ be the standard quotient. By Example 2.42, $\deg \varphi q = 2$ because φq reduces to a homeomorphism from each component of $S^n - S^{n-1}$ (northern and southern hemisphere) to $\mathbb{R}P^n - \mathbb{R}P^{n-1}$ and each homeomorphism can be obtained by the other by precomposing with antipodal map of S^n . Thus $\deg \varphi q = \deg(1) + \deg(-1) = 1 + (-1)^{n+1} = 2$. But φ_* sends 1 to 1, so q_* must send 1 to 2. Thus $f_*(1) = g_*q_*(1) = g_*(2) = 2g_*(1) \implies \deg f$ is even since $g_*(1) \in \mathbb{Z}$.

Thus $\deg f$ is even. □

Problem 2.2.14b. Show that if X is a CW complex then $H_n(X^n)$ is free by identifying it with kernel of cellular boundary map $H_n(X^n, X^{n-1}) \rightarrow H_n(X^{n-1}, X^{n-2})$

Proof. Let c_k be the number of k cells. Note $H_n(X^n, X^{n-1}) \cong H_n(X^n/X^{n-1}) \cong H_n(\bigvee_{i=1}^{c_n} S^n) = \bigoplus_{i=1}^{c_n} H_n(S^n) = \mathbb{Z}^{c_n}$. We have the following long exact sequence of (X^n, X^{n-1}) and (X^{n-1}, X^{n-2}) .

$$\begin{array}{ccccccc} & & & & H_{n-1}(X^{n-2}) = 0 & & \\ & & & & \downarrow & & \\ 0 = H_n(X^{n-1}) & \longrightarrow & H_n(X^n) & \xrightarrow{i_n} & H_n(X^n, X^{n-1}) & \xrightarrow{j} & H_{n-1}(X^{n-1}) \\ & & & \searrow d_n & & \downarrow i_{n-1} & \\ & & & & & & H_{n-1}(X^{n-1}, X^{n-2}) \end{array}$$

Note since $H_n(X^{n-1})$ and $H_{n-1}(X^{n-2})$ are 0, i_{n-1} and i_n are injective so the diagram becomes.

$$\begin{array}{ccccccc} & & & & H_{n-1}(X^{n-2}) = 0 & & \\ & & & & \downarrow & & \\ 0 = H_n(X^{n-1}) & \longrightarrow & H_n(X^n) & \xrightarrow{i_n} & H_n(X^n, X^{n-1}) & \xrightarrow{j} & H_{n-1}(X^{n-1}) \\ & & & \searrow d_n & & \downarrow i_{n-1} & \\ & & & & & & H_{n-1}(X^{n-1}, X^{n-2}) \end{array}$$

By injectivity and exactness, $H_n(X^n) = i_n(H_n(X^n)) = \ker j$. However note $d_n = i_{n-1}j$ and $\ker d_n = j^{-1}(\ker i_{n-1})$ but since i_{n-1} is injective, $\ker i_{n-1} = \{0\}$, $\ker d_n = j^{-1}(\{0\}) = \ker j = H_n(X^n)$. Since $H_n(X^n, X^{n-1}) = \mathbb{Z}^{c_n}$ and $H_n(X^{n-1}, X^{n-2}) = \mathbb{Z}^{c_{n-1}}$ and j is a map of free abelian groups and $H_n(X^n)$ is kernel of j , $H_n(X^n)$ is free. □